

Joseph Zou

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EDUCATION

Vanderbilt University

Nashville, TN

Bachelor of Science in Computer Science (GPA: 3.85/4.00)

Aug. 2021 – May 2025

- **Relevant Courses:** Database Management Systems, Operating Systems, Algorithms, Computer Networks, Artificial Intelligence, Machine Learning, Data Science, Security & Privacy in Pervasive Environments (PhD Level)

EXPERIENCE

Software Engineer Intern

June 2024 – April 2025

Kinetik Care

New York City, NY

- Developed an admin portal using **Retool**, **JavaScript**, and **MongoDB**, integrating company services into a centralized interface, saving the implementation team time and client onboarding by **16%** through automation.
- Collaborated with multiple teams to analyze relationships among stakeholders (transportation providers, brokers, and health plans), delivering actionable insights by enhancing visibility and streamlining logistics operations.

Software Engineer

February 2024 – August 2024

Vanderbilt University Medical Center

Nashville, TN

- Spearheaded the development of a full-stack **React Native** application by architecting the system, leading product design and UI/UX, and integrating data pipelines via the **RedCap API** to securely store group patient data.
- Engineered, trained, and deployed a high-accuracy **YOLOv7** object detection model within a **Docker** container, automating blister pack pill counting from curated datasets, reducing manual review by clinical physicians.

Software Engineer

November 2023 – Present

Mother to Mother

Nashville, TN

- Developed a **Progressive Web App** and admin portal using Prisma ORM, deployed on AWS EC2 with RDS, replacing **10+ years** of spreadsheets and paper records, reducing data entry and retrieval time by **70%**.
- Continued maintaining a mission-critical platform supporting **5,000+ children and families**, streamlining workflows and reporting to save **20+ staff hours weekly**—becoming indispensable to daily operations.

Undergraduate Teaching Assistant

August 2024 – December 2024 and January 2025 – April 2025

Vanderbilt University

Nashville, TN

- **CS 3265: Database Management Systems** and **CS 2201: Program Design and Data Structures**
- Led office hours, Zoom sessions, exam proctoring, and project evaluations in **Python** and **SQL** for 230 students

PROJECTS

PassWin | AWS, React, MongoDB

- Built a zero-knowledge password manager with Argon2id key derivation algorithm and AES-GCM encryption.
- Deployed on AWS EC2 with Nginx reverse proxy, SSL/TLS, MongoDB Atlas, Express, robust user session management and authentication, and GitHub Actions CI/CD, delivering secure, end-to-end encrypted storage.

Image Recognition and Joint Detection for Rock Paper Scissors

- Collaborated with a team to investigate machine learning techniques to classify 5,000 rock paper scissors images
- Reached 99% validation accuracy on base and expanded datasets, using image data and joint detection information
- Designed a rock paper scissors game program to classify real-time hands within 45 centimeters of the camera

TECHNICAL SKILLS

Languages: Java, Python, C++, TypeScript, JavaScript, HTML, CSS, SQL

Frameworks / Libraries: React, Express, Expo, Node.js, Next.js, TensorFlow, Prisma ORM, Mongoose

Tools: Git, Docker, AWS, Vite, Redis, OAuth, MongoDB, PostgreSQL, Retool, Postman